

Ultimate Nutrition

Super Glutamax Capsules

Sore and stiff muscles are not uncommon after high-intensity workouts. It may sometimes happen that despite regular exercise to keep in good health, athletes may become more prone to catching colds or other illnesses. This should serve as a red flag that your muscles are under a great deal of stress, resulting in muscle damage and soreness that may also indicate a compromised immune system.

Although it is not possible to completely reverse muscle damage due to exercise, research has shown that the damage due to exercise can be considerably minimized. Equally important, various studies suggest that nutritional measures can be taken to boost your immune system, which may become weakened by exercise stress. Glutamine is the almost "wonder nutrient" that can help manage muscle soreness, boost energy levels, help the body to secrete its own growth hormone, and enhance immunity.

The amino acid glutamine is a great source of energy for white blood cells and other cells of the immune system. Low levels of glutamine, therefore, may weaken the immune system. Also, glutamine plays a critical role in gastrointestinal health. While it is largely made in the lungs and the brain, glutamine is also present in muscle and liver cells. Under stress, including physical stress of exercise, the amount of glutamine in the muscle is decreased dramatically. Therefore, there is a corresponding decrease in the blood levels of glutamine. If the body cannot keep up with the demand to produce this amino acid in the required amounts, glutamine must be supplemented from dietary sources.

A strict, strenuous training regimen that does not allow sufficient time for recovery may cause overtraining syndrome (OTS). This syndrome presents itself as decreased performance, depressed mood and increased rate of infections, and runners, bicyclists, skiers and, among others, rowers are known to suffer from it. This is due to persistent low levels of glutamine in the blood plasma of these athletes. Glutamine supplementation helps athletes within as short a period of time as a week.

Glutamine is also known to induce the secretion of growth hormone. Its effect on the release of growth hormone depends upon a regular exercise regimen. Growth hormone is an extremely important hormone, which triggers the release of IGF-1 (insulin growth factor-1) that is the workhorse to help reduce weight, increase lean body mass, increase bone density and, of course, promotes general health and wellbeing considerably.

Ultimate Nutrition's' Glutamax is designed to deliver these benefits of glutamine. It helps reduce the impact of overtraining syndrome, promotes faster muscle recovery after intense and regular exercise, it provides "fuel" for the immune system for production of white blood cells to ward off infections, and induces growth hormone making use of the body's own supply of this vital hormone. Therefore, Glutamax is the choice of athletes to nutritionally strengthen their bodies with this wholesome amino

acid.

SELECTED REFERENCES

- Walsh et al., "Glutamine, Exercise, and Immune Function," Sports Med.: 26, 177, 1998
Rowbottom et al., "The Emerging Role of Glutamine as an Indicator of Exercise Stress and Overtraining," Sports Med.: 21, 80, 1996
Rohde et al., "Glutamine, Exercise and the Immune System - Is There a Link?" Exer. Immunol. Rev.: 4, 49, 1998
Newsholme, E., "A Biochemical Mechanism to Explain Some Characteristics of Overtraining," Adv. Nutr. & Topics Sports: 32, 79, 1991
Hiscock, N. & Mackinnon, L., "A Comparison of Plasma Glutamine Concentrations in Athletes of Various Sports," Med. Sci. in Sports Exer.: 30, 1693, 1998

FREQUENTLY ASKED QUESTIONS

1. Why Take Supplemental Glutamine When it is Present in Many Foods?

It is true that glutamine occurs naturally in a wide variety of foods. It does, however, lose its biological activity by cooking. For example, spinach and parsley are good sources of glutamine when eaten raw.

2. Isn't Glutamine Contained in Nutritional Supplements?

Yes, it is! Since the benefits of glutamine are not widely known by manufacturers of sports nutrition products, small amounts of glutamine are included in everything from protein powders to replacement meals. The amount of glutamine in these products is not sufficient, however, to derive benefits from it.

3. How Much Glutamine Should be Taken To Secrete Growth Hormone?

In one study published in 1995 (Welborne, American Journal of Clinical Nutrition), it was reported that two grams of glutamine per day in three equal doses may suffice to prime the body to induce growth hormone secretion. Of course, the amount of glutamine needed to secrete growth hormone depends on age, the state of general health, lifestyle (sedentary vs. active) and a host of other factors. An updated review on growth hormone secretion and its health benefits may be found in Dr. Aftab Ahmed's "Health Benefits of Growth Hormone," Keats Publishing, 1998.

4. What is the Optimal Dose of Glutamine for Athletes?

There is no cookie-cutter dosage recommendation for everyone. The dosage should be determined in consultation with the input from the trainer and/or healthcare provider. It is also essential to keep the objectives in mind. If the objective is to replenish glutamine lost during regular exercise, the amounts needed may not be large. Many people involved in endurance and high-intensity training take large dosages of glutamine. This, however, should be determined by experimentation and the dosage escalated gradually in consultation with the primary care physician.

5. Are There Any Side Effects?

Taken in large amounts in one sitting may cause dry mouth, nausea, dizziness and intense discomfort.